

Software issues

mystudio related

Q: Details about downloading firmware for mystudio

1. Generally, the firmware is already built-in when leaving the factory. Unless the machine is abnormal, there is no need to re-flash the firmware
2. You need to connect to the network during the firmware download process
3. Select the firmware for the corresponding model, and do not choose the wrong model

python related

Q: The running prompt shows that the library file is missing Q: The error message: ModuleNotFoundError: No module named "pymycobot", how to deal with it?

- A: The reason for the error may be that pymycobot is abnormal. The corresponding solution is to reinstall pymycobot. The command is `pip3 install pymycobot --upgrade --user`

Q: Is there an API for AGV to control distance or angle?

- A: The distance control in the current AGV API is determined by a combination of time and speed calculations. There is currently no API specifically for specifying the moving distance or wheel angle. If you need to travel a specific distance, you can calculate the time and speed required for distance conversion.

Q: How to solve the problem of missing scripts folder when running the lidar file on AGV JN?

- A: Since the AGV JN image version is relatively old, this problem cannot be solved by re-burning the image You need to download the missing source files from the GitHub file and import them into AGV The specific link is as follows: https://github.com/elephantrobotics/myagv_ros/tree/myagv_ros_2023JN/myagv_odometry

ROS related

Q: How to deal with the failure to start the radar odometer or abnormal map building?

- A: Enter the following two commands to enable the radar

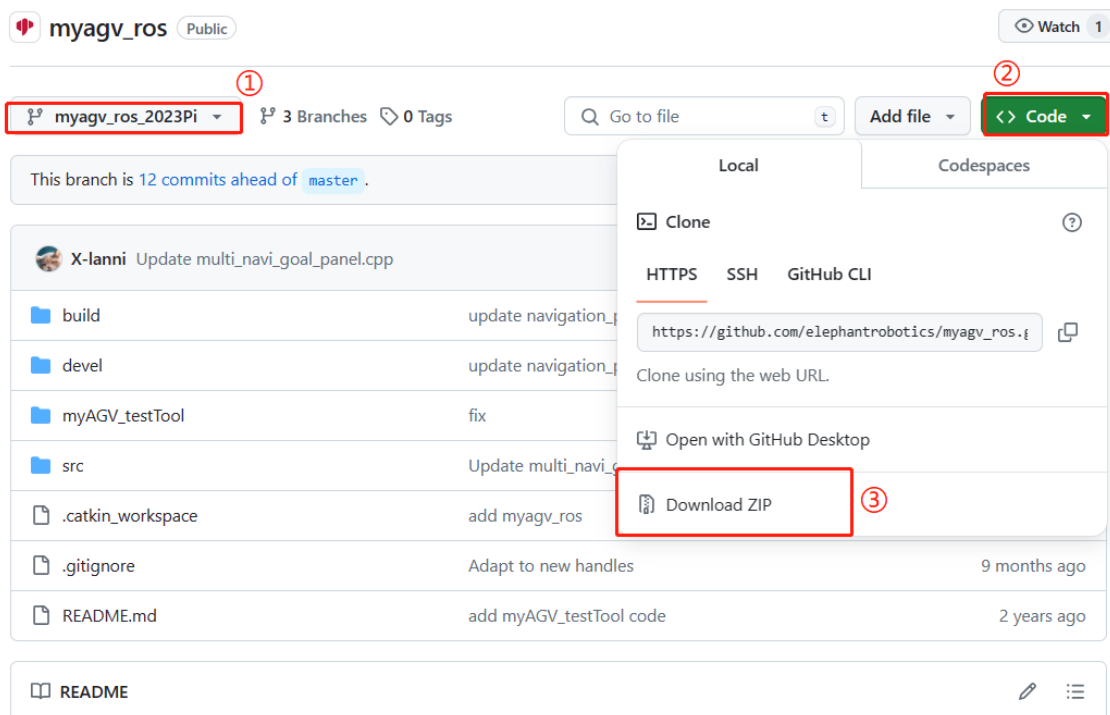
```
//Jump to the laser radar startup directory
```

```
cd myagv_ros/src/myagv_odometry/scripts
```

```
//Power on to enable the radar. After powering on, the radar will send data through the serial port
```

```
./start_ydlidar.sh
```

If there is no start_ydlidar.sh file in the system's built-in myagv_ros package, please re-download the myagv_ros package. The command is: git clone https://github.com/elephantrobotics/myagv_ros/tree/myagv_ros_2023Pi Or after selecting the corresponding branch, click "Code" and then click "Download ZIP" to download the ros package to the local computer and then use a USB flash drive to put the ros package in the AGV system.



If the model is AGV-JN, please switch to the AGV-JN branch

myagv_ros Public Watch 1

myagv_ros_2023JN 3 Branches 0 Tags Go to file Add file Code

This branch is 20 commits ahead of master.

Commit	Message	Time
X-lanni	Update multi_navi_goal_panel.cpp	6ee6e70 · last month 32 Commits
build	fix cmake errors with rtabmap and orbbecSDK	4 months ago
devel	fix cmake errors with rtabmap and orbbecSDK	4 months ago
myAGV_testTool	fix	2 years ago
src	Update multi_navi_goal_panel.cpp	last month
.catkin_workspace	add myagv_ros	2 years ago
.gitignore	update teleop and navigation files	5 months ago
README.md	add myAGV_testTool code	2 years ago

Notes:

1. After the replacement, you need to delete the myagv_ros folder in the original home/er directory, unzip the newly downloaded ROS source code package and put it in the home/er directory and change the name to myagv_ros, that is, completely replace the myagv_ros source code
2. After completing the replacement of myagv_ros, you need to compile it. The corresponding operation is to enter the command in the myagv_ros directory: catkin_make to compile. After the compilation shows 100% progress, the radar mapping function can be started normally.

Q: Why does the AGV circle when navigating? Is this normal?

AGV's circle behavior currently exists and is normal. There are two main reasons for the circle behavior:

1. The car's odometer is not accurate enough. If the error is sent midway, it needs to circle, calculate the error, and correct the positioning
2. The tolerance value of the car's navigation target point is too low. Although the target point has been reached on the map, the car thinks that it is a little bit away from the destination point, so it will circle around the target point.

The factory has set a suitable tolerance value, but the user can try to modify the tolerance value. The method of modifying the file parameters refers to the figure below. The two parameters circled in the figure below can be increased in increments of 0.05

[myagv_ros / src / myagv_navigation / param / planner.yaml](#)

 X-lanni update param and multipoint_navigation.launch

Code

Blame

Executable File · 99 lines (81 loc) · 3.9 KB

```
1 controller_frequency: 5.0
2 recovery_behaviour_enabled: true
3
4 NavfnROS:
5   allow_unknown: true # Specifies whether or not to allow navfn to create plans that traverse unknown space.
6   default_tolerance: 0.1 # A tolerance on the goal point for the planner.
7
8 TrajectoryPlannerROS:
9   # Robot Configuration Parameters
10  acc_lim_x: 2
11  acc_lim_theta: 0.5 #2
12  acc_lim_y: 0.0
13
14  max_vel_x: 0.2
15  min_vel_x: 0.01
16
17  max_vel_theta: 0.1 #0.5
18  min_vel_theta: -0.1 #-0.5
19  min_in_place_vel_theta: 0.25
20
21  holonomic_robot: false
22  escape_vel: 0.0
23
24  # Goal Tolerance Parameters
25  yaw_goal_tolerance: 0.25
26  xy_goal_tolerance: 0.25
27  trans_stopped_vel: 0.05
28  theta_stopped_vel: 0.01
```

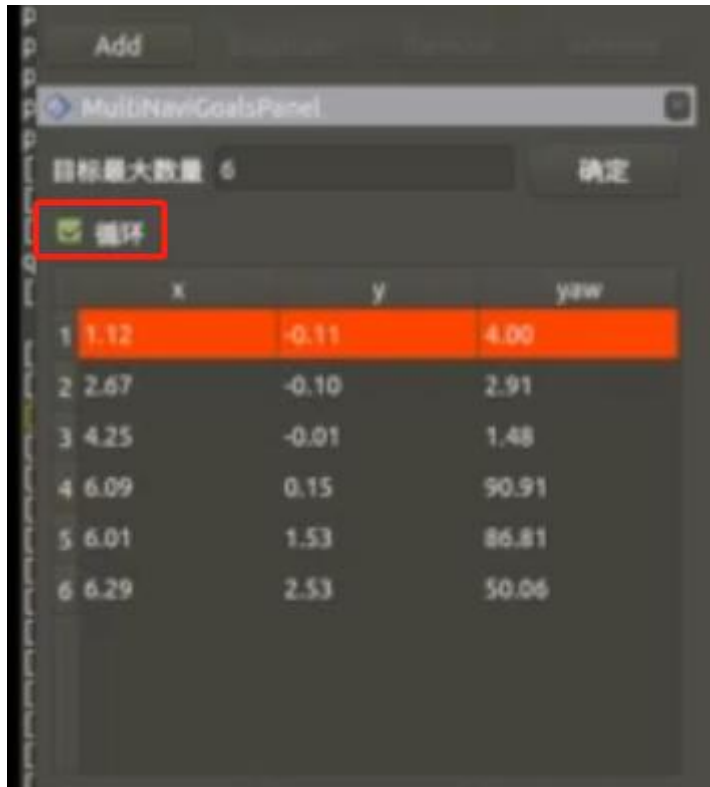
There is currently no better solution for this kind of circle behavior. When the AGV circles and cannot continue to the next positioning point, please try to manually push the AGV to change the position of the AGV so that the AGV can get closer to the navigation point.

Q: The Raspberry Pi and Jetson Nano machines are not very smooth when using ROS. Sometimes they are stuck. Is there any optimization method?

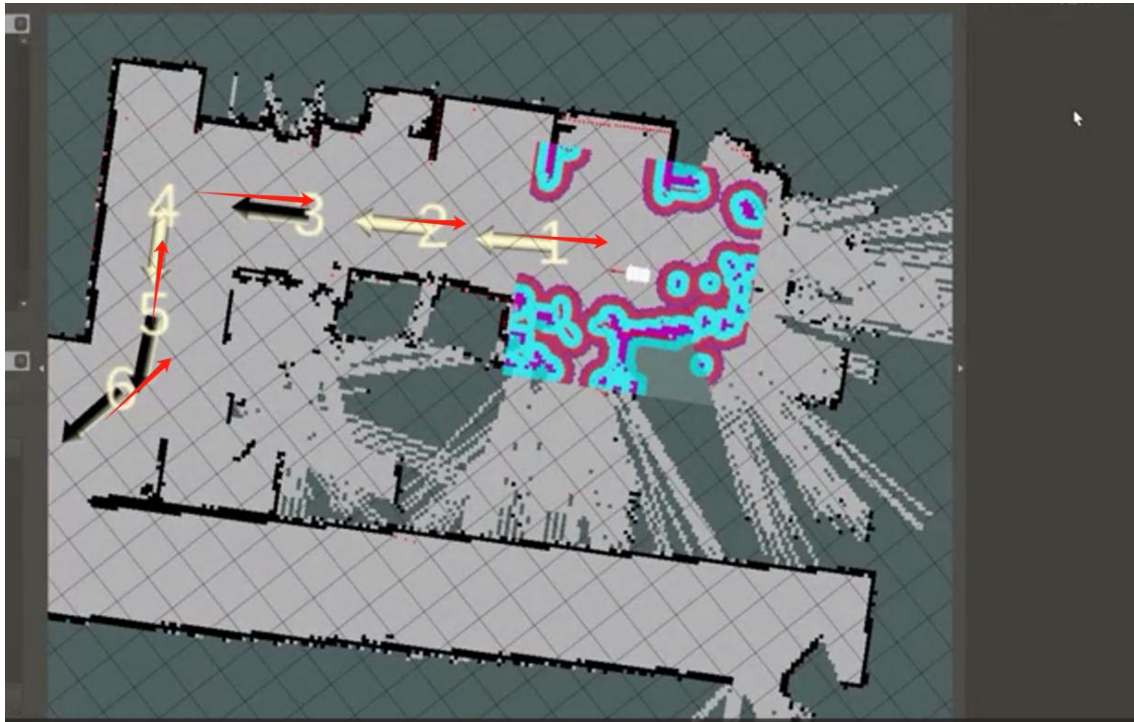
Although Raspberry Pi and Jetson Nano are powerful small computers, due to their limited hardware configuration, it is normal to have stuck when running some tasks that require a lot of computing resources. For example, when running ROS development, if it involves tasks that require a lot of computing resources such as image recognition, the performance may not meet the requirements, resulting in stuck phenomena. This problem is largely caused by insufficient hardware performance and insufficient computing power. You need to optimize and improve it according to the specific situation. It is recommended to try to use the following methods to optimize:

- ① Optimize code: By optimizing the code, reduce unnecessary calculations, improve the execution efficiency of the code, and thus reduce the burden on Raspberry Pi and Jetson Nano.
- ② Upgrade hardware: You can consider using a higher-performance Raspberry Pi model, or other hardware platforms with better performance and higher computing power, such as NVIDIA NX series motherboards.
- ③ Distributed computing: Tasks can be distributed to multiple devices for processing, thereby reducing the burden on a single device and improving the performance of the entire system.

Q: How to deal with the deviation problem of AGV using radar multi-point navigation?



There are two reasons for the deviation of multi-point navigation: ① The AGV odometer is not accurate enough. After running multiple points, the offset has accumulated. Some of it can be corrected by AMCL, but not all of it. The problem exists and cannot be solved for the time being. It is necessary to wait for the optimization of the mileage calculation method or the upgrade of the hardware. ② Improper navigation operation. If the loop navigation mode is set, after setting multiple points and running the last point, the car will run to the first point and complete a cycle. However, due to the error of the wheel odometer, the path cannot be planned on the return trip. The car may run directly from the last point to the second point and fail to reach the first point. Therefore, it is recommended to add more points to complete the cycle when using the loop, as shown in the figure below, and try to form a closed loop.



****Q:** What is the input source of the AGV mapping algorithm? Does cartographer support it?

A: The cartographer mapping algorithm is still under development and there is no exact launch time.

Q: What is the input of the radar mapping algorithm?

A: The gmapping mapping algorithm requires a lidar as input.

The rtabmap mapping algorithm requires a lidar and a 3d camera as input.

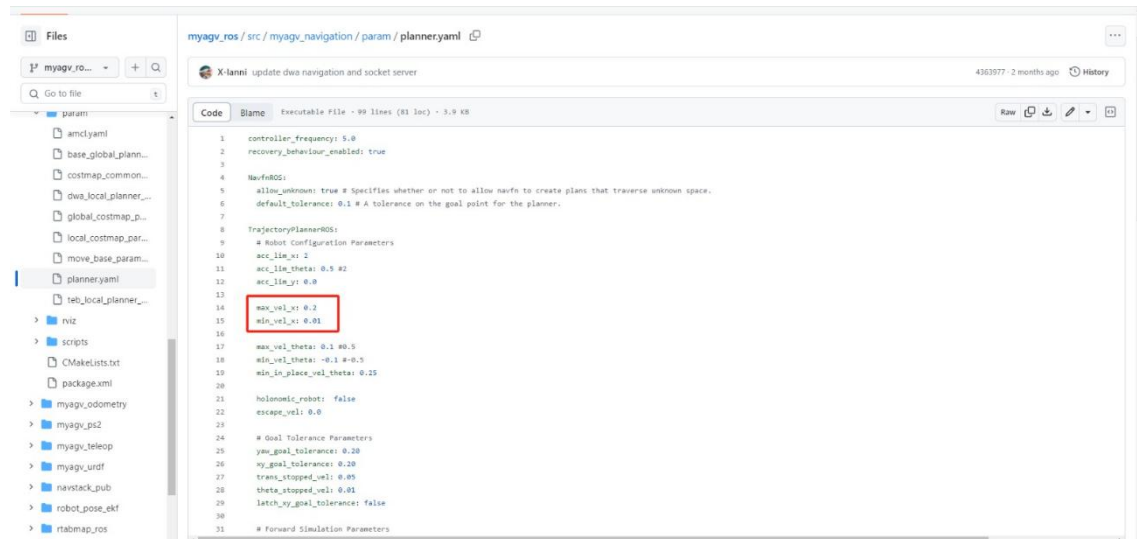
Q: How to use the ordinary camera of the car as the input algorithm

A: There is currently no case of using a 2d camera as input. It is recommended to check the usage of ORB-SLAM. Users need to study it themselves. The current development of myagv on visual SLAM is based on 3d cameras. Currently, the camera in front of myagv is only used to recognize aruco codes.

Q: Can the speed of the car in the mapping algorithm be adjusted?

A: The speed during navigation can be adjusted, but it is not recommended because the speed parameter is uncertain whether it will affect the overall

navigation effect The circle in the figure below is the maximum and minimum forward speed during navigation. If you want to go faster, just increase 0.2 a little bit



```

1 controller_frequency: 5.0
2 recovery_behavior_enabled: true
3
4 NavfnROS:
5   allow_unknown: true # Specifies whether or not to allow navfn to create plans that traverse unknown space.
6   default_tolerance: 0.1 # A tolerance on the goal point for the planner.
7
8 TrajectoryPlannerROS:
9   # Robot Configuration Parameters
10  acc_lim_x: 2
11  acc_lim_theta: 0.5 #2
12  acc_lim_y: 0.8
13
14  max_vel_x: 0.2
15  min_vel_x: 0.01
16
17  max_vel_theta: 0.1 #0.5
18  min_vel_theta: -0.1 # -0.5
19  min_in_place_vel_theta: 0.25
20
21 holonomic_robot: false
22 escape_vel: 0.0
23
24 # Goal Tolerance Parameters
25 yaw_goal_tolerance: 0.20
26 xy_goal_tolerance: 0.20
27 trans_stopped_vel: 0.05
28 theta_stopped_vel: 0.01
29 latch_xy_goal_tolerance: false
30
31 # Forward Simulation Parameters
  
```

Q: Why do some points set in the middle appear to be skipped during radar multi-point navigation?

A: There are two reasons and optimization methods ① The matching degree of the established map is not high. It is recommended to let the AGV run a few more laps during mapping to allow the radar to fully scan and make the mapping accurate ② During multi-point navigation, the navigation distance between the front and rear points is too long, the positioning deviation is relatively large, and the path planning difficulty increases. If the navigation fails, the corresponding point will be skipped and navigate to the next point. Therefore, it is recommended to add a few more navigation points before the starting point and the target point.

Hardware problem

Q: What is the model of AGV radar? Can you provide the performance parameters of the radar?

- A: YDLIDAR X2L About fov horizontally is 0-360°, vertically is 0-1.75° 4000 points, 4000 times per second

表1 YDLIDAR X2L 性能参数

项目	最小值	典型值	最大值	单位	备注
测距频率	/	4000	/	Hz	每秒测距 4000 次
扫描频率	8	10	12	Hz	PWM 或电压调速
测距范围	0.12	/	>8	m	室内环境, 80%反射率物体
扫描角度	/	0-360	/	Deg	/
绝对误差	/	2	/	cm	测距≤1m 时
相对误差	/	3.5%	/	/	1m<测距≤6m
俯仰角	0.25	1	1.75	Deg	/
角度分辨率	0.72 (8Hz)	0.90 (10Hz)	1.08 (12Hz)	Deg	/

注 1: 为工厂 FQC 标准值, 80%反射率材质物体。

Q: Where is the installation method of the 7-inch screen? What to do if it cannot display?

Please refer to the link for screen installation and connection method: AGV installation

screen: <https://docs.qq.com/doc/DU1p0VnBCRkFIQk9J> <https://docs.qq.com/doc/DU2xWd29ldkFHcFNR>

Q: Does AGVJN have a built-in network card?

A: Yes, it has a built-in wireless network card

Q: Can AGV be used as a composite robot with myArm300pi? How?

A: The steps to start the AGV and myarm300 composite robot are as follows:

1. Start the Server.py file in myarm, Download path: <https://github.com/elephantrobotics/pymycobot/blob/main/demo/Server.py> Note: The Server.py file is required and the baud rate 1000000 in the file is changed to 115200
2. Start the Client.py file in myAGV. The control code in Client.py supports customization, for example

```
from pymycobot import MyArmSocket
```

```
import time
```

```
mc = MyArmSocket("192.168.10.97",9000)
```

```
for i in range(3):
```

```
mc.send_angles([0, 0, 0, 0, 0, 0, 0], 50)
```

```
print(mc.get_angles())
```

```
time.sleep(2)
```

```
mc.send_angles([90, 0, 0, 0, 0, 0, 0], 50)
```

```
print(mc.get_angles())
```

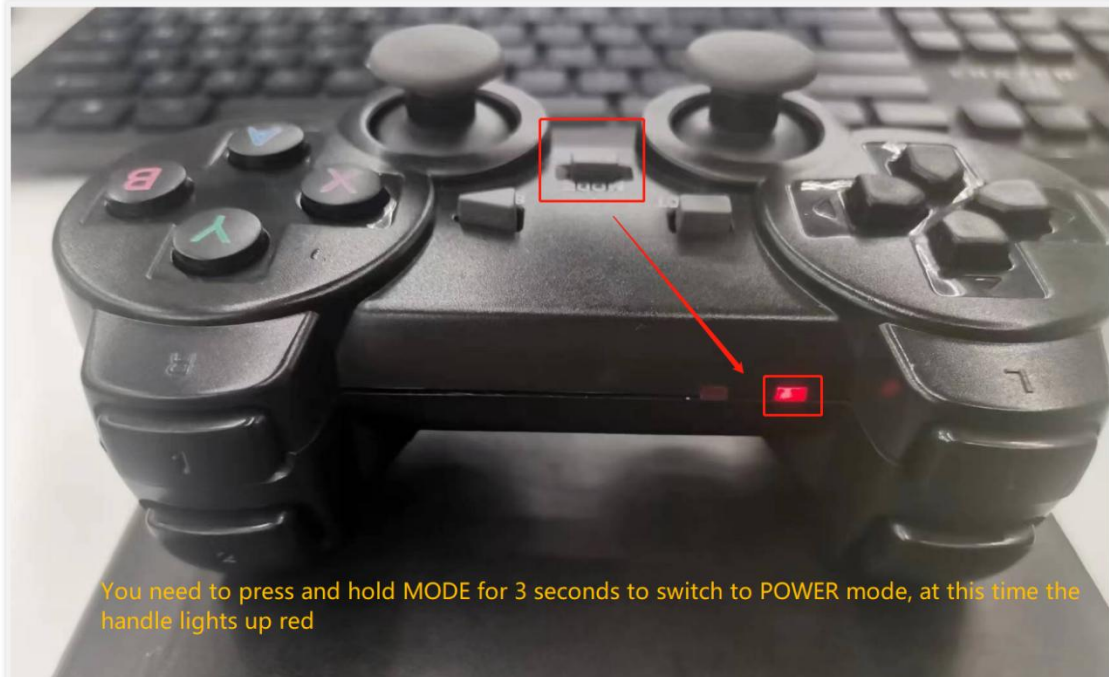
```
time.sleep(2)
```

Q: Can AGV use other 3D cameras, such as D435? How should I find its launch file?

A: Currently, Intel's D435 camera has not been adapted. Now there is only Obbec's. If you want to use this Intel camera, it is recommended to refer to the development manual of the 3D mapping algorithm and develop it yourself

Q: How to deal with abnormal use of AGV handle?

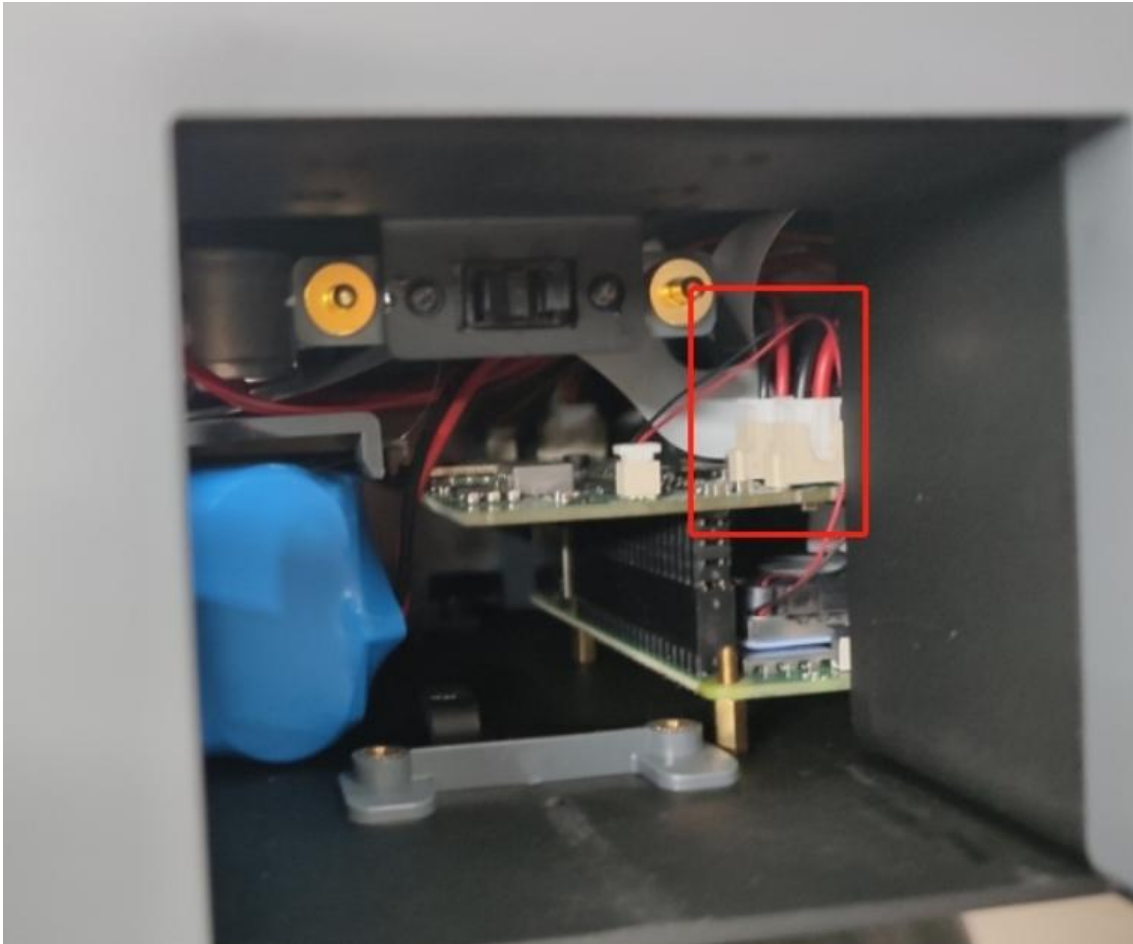
第二步需要长按3秒MODE，切换成POWER模式，此时手柄亮红灯

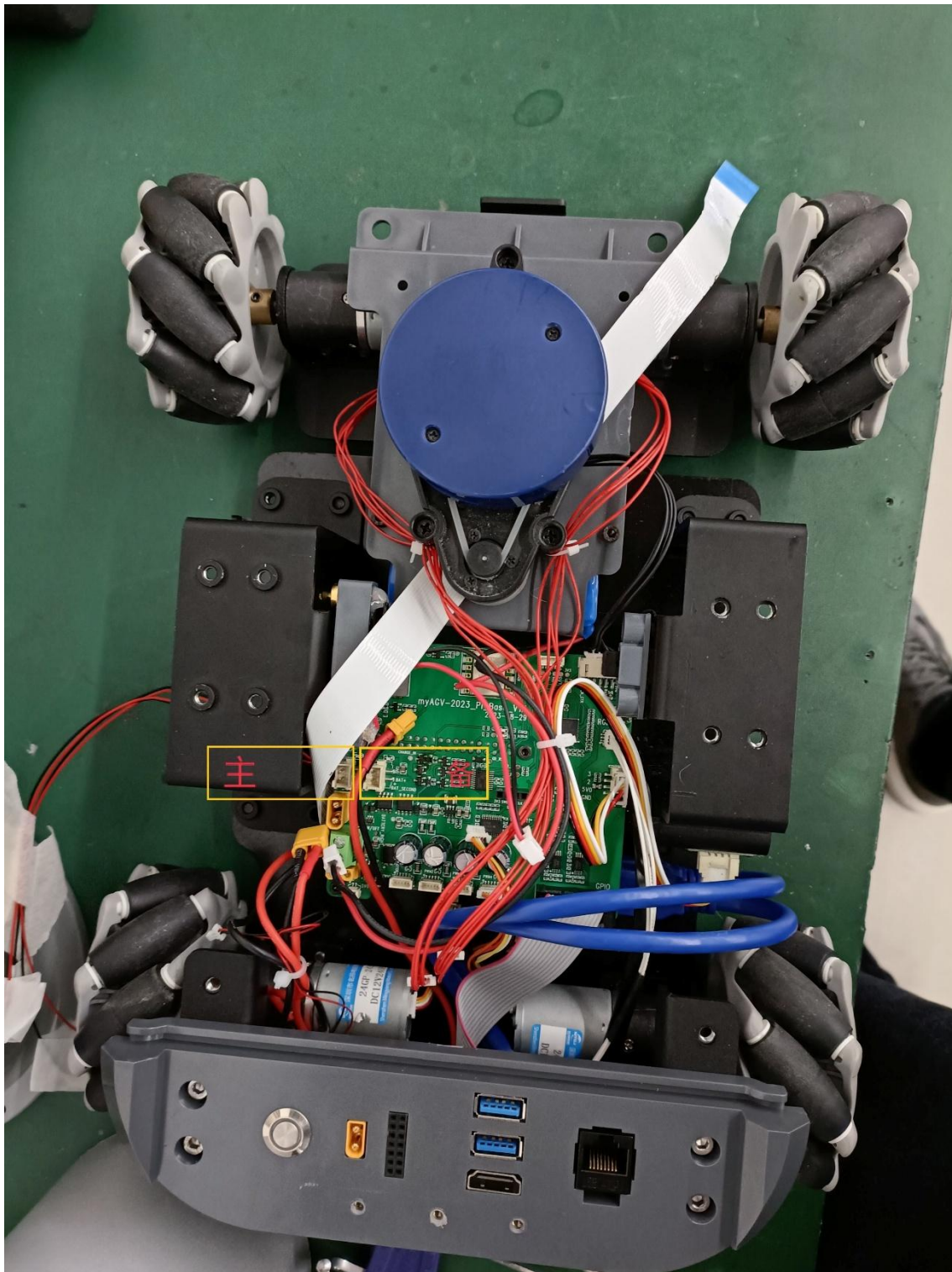


Q: What should I do if the charger cannot be charged?

1. Please confirm that you are using the power adapter we provide
2. Can you tell me whether the reason why this AGV cannot be charged for one hour is that after the matching screen is connected, the screen does not display any system screen or the power button light does not light up after the AGV power switch is turned on?
3. If the AGV power light is on, have you tried to replace the HDMI screen to see if you can enter the AGV system and further check whether the AGV motor can be controlled normally?
4. If the AGV power light is not on, can you open one of the side hatches of the AGV after connecting the power adapter and take a picture of the inside for me to see? I need to see if there is an indicator light inside
5. When the AGV is charging for 1 hour, are you sure that the power button of the AGV is turned off? When the power button is pressed, that is, the AGV is started, it is used while charging, and the charging effect will be relatively poor. So when charging, the power button switch needs to be turned off. Is this how you charge it?

6. The two terminals circled in the picture below are the key to charging connection. You need to check if any cables are disconnected. You can try to unplug and replug these two terminals.





Q: Is there any way to check if the battery is fully charged?

A: After charging for about 1 hour, you can turn on the system and check the voltage and current values in the visual UI

Q: Can the main battery and the backup battery be charged together? Can the backup battery be charged on the car?

A: Put the backup battery in the battery compartment. It will start charging when it is below 11V. The main battery will be charged first, and then the backup battery

Q: What should I do if the serial port number `/dev/ttyACM0` cannot be recognized?

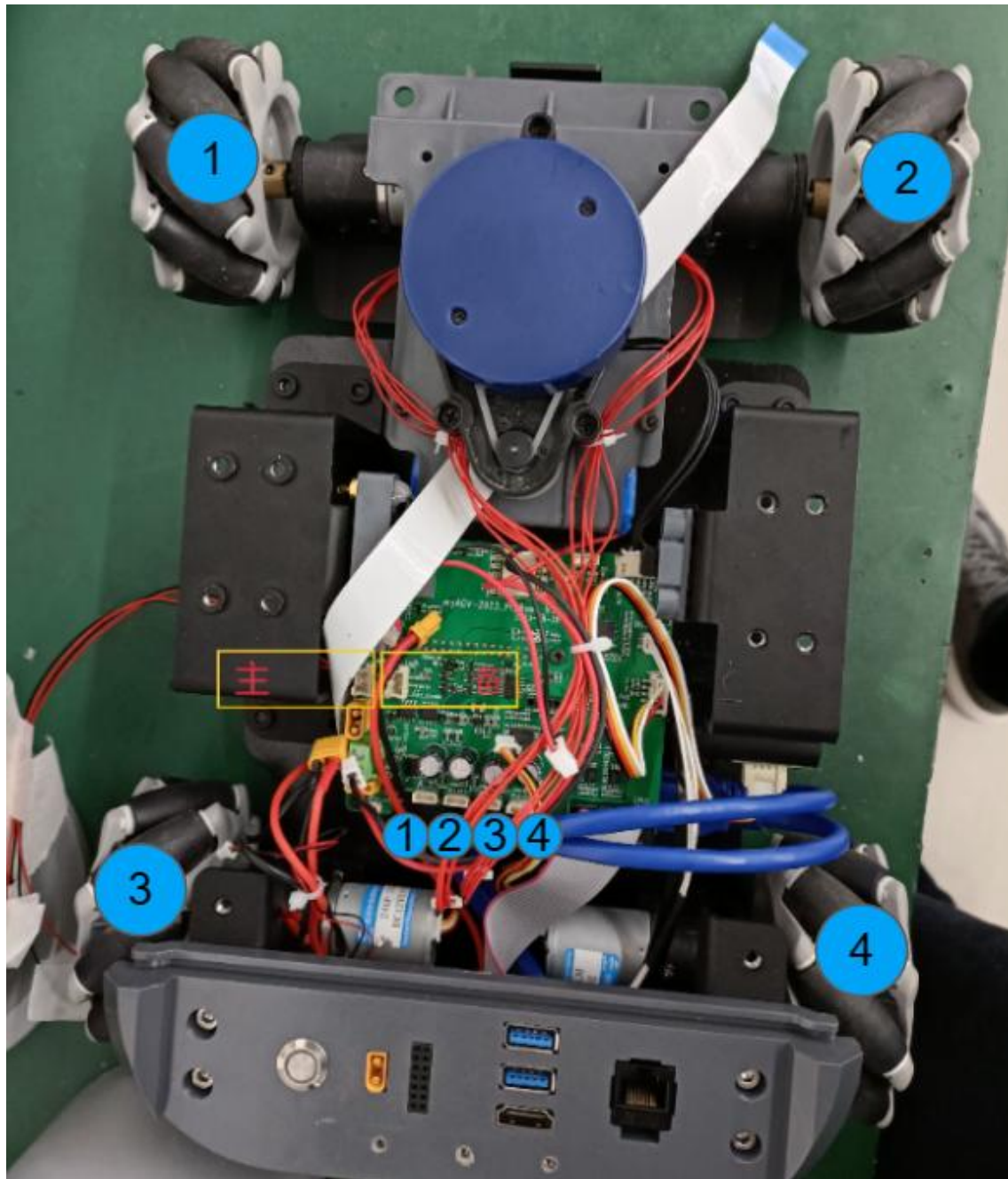
Disassemble the right rear wheel, reinstall the serial port cable, and then use the command `ls /dev/tt*` to check whether it is recognized normally. Note that you cannot pull out the serial port cable by force. You should gently lift the plastic sheet and then remove the cable. You can refer to the removal of the csi cable.



Q: Which is the motor control pin?

Please refer to the following 4 groups of pins, 2 of which control 1 motor, but we only provide integrated motor control interface. We cannot provide support for the use of pins, and you need to do secondary development yourself. 32 33 25 26 27 14 18 23

Q: Can you provide the line sequence corresponding diagram of the 4 motors of the car and the motor interface of the driver board?



Q: How to replace the expansion board of the car?

A: Refer to the video: <https://drive.google.com/file/d/1NYZTkAqMKsrFyl1-HzsuKRmJr6ksMpvL/view?usp=sharing>

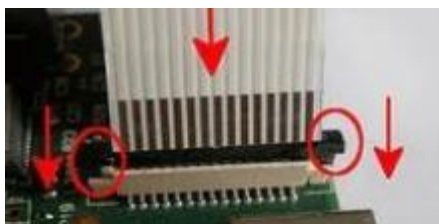
Q: Is there anything I need to pay attention to when disassembling and assembling the CSI camera?

A: Please note that when installing or removing the CSI ribbon cable, please gently pull or push the plastic sheet, and do not pull the CSI ribbon cable directly.

Removal of the CSI ribbon cable



Installation of the CSI ribbon cable



Q: What is the power of the myCobot Pro 600 robot arm when it is working?

- A: 240W 48V 5A

Q: Where is the base coordinate origin of 600/630?

The base coordinate origin of 600/630 is at the position of ② (as shown in the figure), not at the position of ①

Q: Re-zero calibration in cmd: this is not necessary, (Encoder abnormality troubleshooting-600)

- A: It may be because the motor encoder has an error due to long-term non-use. Now please disassemble the housing of manipulator J2 and J3, check whether there is a red light in the encoder, find the encoder line, press it, and the encoder will have a green light, and it can be used normally.

Q: What is the 630 master control? What is the native system?

- A: Raspberry Pi 4B 8G memory debian system

Q: Why does the Raspberry Pi cause the phenomenon of being unable to enter the operating system?

- A: There are two possibilities. The first is that the instantaneous current causes the TF card and the Raspberry Pi system image to be damaged. It may also be that the TF card capacity is insufficient and the papermaking system is hung up. Because the factory memory left for customers is not much, most of it is occupied by the mycobot_ros folder. Customers can delete the folders except for the model product before the product has a black screen and prompts insufficient memory to reserve product usage space. If there is too much memory data, it cannot be turned on when it is full.